

SIMATS ENGINEERING

Name	Register Number	Course Code	Course
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36. Recognise watch from the given image by general Object recognition using OpenCV.

AIM:

Recognise watch from the given image by general Object recognition

PROGRAM:

```
import cv2

# Read the image
img = cv2.imread(r"C:\Users\Susmitha\Downloads\ITA0510 Lab\exp3.png")

if img is None:
    print("Error: Image not found!")
    exit()

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Reduce noise
blur = cv2.GaussianBlur(gray, (5, 5), 0)

# Detect edges
edges = cv2.Canny(blur, 50, 150)

# Find contours
contours, _ = cv2.findContours(edges, cv2.RETR_EXTERNAL,
cv2.CHAIN_APPROX_SIMPLE)

# Find the largest contour (assumed to be the watch)
largest_contour = max(contours, key=cv2.contourArea)

x, y, w, h = cv2.boundingRect(largest_contour)

# Draw bounding rectangle
cv2.rectangle(img, (x, y), (x + w, y + h), (0, 255, 0), 2)

# Label the object
```

```
cv2.putText(img, "Watch",  
            (x, y - 10),  
            cv2.FONT_HERSHEY_SIMPLEX,  
            0.8,  
            (0, 255, 0),  
            2)
```

```
# Display the result
```

```
cv2.imshow("Recognized Watch", img)
```

```
# Save the result
```

```
cv2.imwrite(r"C:\Users\Susmitha\Downloads\ITA0510 Lab\Recognized_Watch.jpg", img)
```

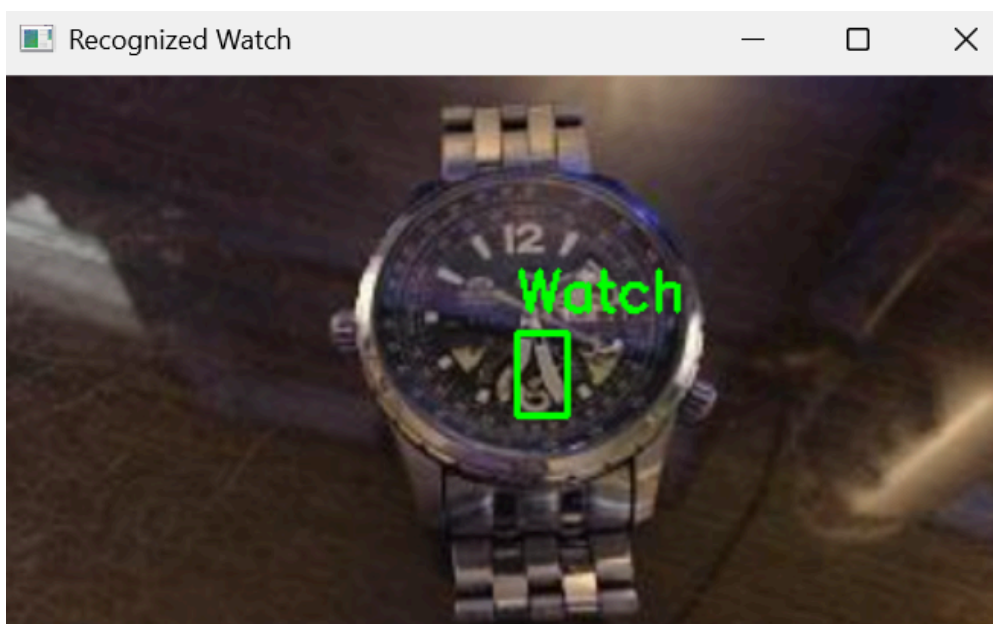
```
cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

INPUT:



OUTPUT:



RESULT:

Thus the program is executed successfully.